

UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES
CENTRO DE CIÊNCIAS EXATAS E TECNOLÓGICAS – CCET
DEPARTAMENTO DE CIÊNCIAS DA COMPUTAÇÃO – DCC

ATIVIDADE AVALIATIVA I

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG
MARÇO/2026

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ATIVIDADE AVALIATIVA I

Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Geometria Analítica e Álgebra Linear do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

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• Exercício 1.1 - Impares -

1. a - $x_1 + 5x_2 + \sqrt{2}x_3 = 1 \rightarrow$ É uma equação linear.

b - $x_1 + 3x_2 + x_1x_3 = 2 \rightarrow$ Não é uma equação linear, pois tem multiplicação de variáveis.

c - $x_1 - 7x_2 + 3x_3 \rightarrow$ É uma equação linear.

d - $x^2 + x^2 + 8x_3 = 5 \rightarrow$ Não é uma equação linear, pois duas das variáveis tem expoente diferente de 1.

e - $x_1^{3/5} - 2x_2 + x_3 = 4 \rightarrow$ Não é uma equação linear, pois uma variável está elevada de uma raiz.

f - $11x_1 - \sqrt{2}x_2 + \frac{1}{3}x_3 = 7^{1/3} \rightarrow$ É uma equação linear.

3. a - $\begin{cases} 2x_1 - x_4 = 5 \\ -x_1 + 5x_2 + 3x_3 - 2x_4 = -1 \end{cases} \rightarrow$ É um sistema linear.

b - $\begin{cases} \sin(9x_1 + x_3) = \sqrt{5} \\ e^{-2x_2 - 2x_4} = 1/x^2 \\ 4x_4 = 4 \end{cases} \rightarrow$ Não é um sistema linear pois tem duas equações com funções transcendentais nas variáveis.

c - $\begin{cases} 7x_1 - x_2 + 2x_3 = 0 \\ 9x_1 + x_2 - x_3x_4 = 3 \\ -x_1 + 5x_2 - x_4 = -1 \end{cases} \rightarrow$ Não é, pois uma das equações tem multiplicação de variáveis.

d - $x_1 + x_2 = x_3 + x_4 \rightarrow$ Não é, pois seriam necessárias pelo menos duas equações lineares para formar um sistema linear.

5. $\begin{cases} 7x_1 - x_4 = 5 \\ -x_1 + 5x_2 + 3x_3 - 2x_4 = -1 \end{cases} \rightarrow$ Sim. Como tem 2 equações e 4 incógnitas, 2 incógnitas terão infinitas soluções, tornando o sistema um SPI, com infinitas soluções.

$$7. \begin{cases} 2x_1 - 4x_2 - x_3 = 1 \\ x_1 - 3x_2 + x_3 = 1 \\ 3x_1 - 5x_2 - 3x_3 = 1 \end{cases} \Rightarrow \begin{bmatrix} 2 & -4 & -1 & 1 \\ 1 & -3 & 1 & 1 \\ 3 & -5 & -3 & 1 \end{bmatrix}$$

a. $(3, 1, 1) =$
 $2 \cdot 3 - 4 \cdot 1 - 1 = 1 \quad \checkmark$

É uma solução

$3 - 3(1) = 1 \quad \checkmark$

$3 \cdot 3 - 5(1) - 3(1) = 1 \quad \checkmark$

b. $(3, -1, 1) =$
 $2 \cdot 3 - 4 \cdot (-1) - 1 = 9 \quad \times$

Não é uma solução

c. $(13, 5, 2) =$
 $2 \cdot 13 - 4 \cdot 5 - 2 = 4 \quad \times$

Não é uma solução

d. $(\frac{13}{2}, \frac{5}{2}, 2) =$

É uma solução

$2 \cdot \frac{13}{2} - 4 \cdot \frac{5}{2} - 2 = 1 \quad \checkmark$

$\frac{13}{2} - 3 \cdot \frac{5}{2} + 2 = 1 \quad \checkmark$

$3 \cdot \frac{13}{2} - 5 \cdot \frac{5}{2} - 3 \cdot 2 = 1 \quad \checkmark$

e. $(17, 7, 5) =$ É uma solução

$2 \cdot 17 - 4(7) - 5 = 1 \quad \checkmark$

$17 - 3(7) + 5 = 1 \quad \checkmark$

$3 \cdot 17 - 5 \cdot 7 - 3 \cdot 5 = 1 \quad \checkmark$

9. a. $7x - 5y = 3 \Rightarrow x = \frac{3}{7} + \frac{5}{7}y$

$y = 9$

$S = (\frac{3+5y}{7}, y)$

b. $-8x_1 + 2x_2 - 5x_3 + 6x_4 = 1$

$2x_2 = 1 + 8x_1 + 5x_3 - 6x_4$

$S = (a, \frac{1+8a+5b-6c}{2}, b, c)$

$x_2 = \frac{1+8x_1+5x_3-6x_4}{2}$

2

$x_1 = a$

$x_3 = b$

$x_4 = c$

$$11-a. \begin{bmatrix} 3 & 0 & 0 \\ 3 & -4 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{cases} 3x = 0 \\ 3x - 4y = 0 \\ 0 - y = 1 \end{cases}$$

$$b. \begin{bmatrix} 3 & 0 & -2 & 5 \\ 7 & 1 & 4 & -3 \\ 0 & -2 & 1 & 7 \end{bmatrix} = \begin{cases} 3x + 0 - 2y = 5 \\ 7x + 1y + 4z = -3 \\ 0 - 2y + 1z = 7 \end{cases}$$

$$c. \begin{bmatrix} 7 & 2 & 1 & -3 & 5 \\ 1 & 2 & 4 & 0 & 1 \end{bmatrix} = \begin{cases} 7x + 2y + 1z - 3t = 5 \\ 1x + 2y + 4z + 0 = 1 \end{cases}$$

$$d. \begin{bmatrix} 1 & 0 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{bmatrix} = \begin{cases} x = 7 \\ y = -2 \\ z = 3 \\ t = 4 \end{cases}$$

$$13-a. \begin{cases} -2x_1 = 6 \\ 3x_1 = 8 \\ 9x_1 = -3 \end{cases} \Rightarrow \begin{bmatrix} -2 & 6 \\ 3 & 8 \\ 9 & -3 \end{bmatrix}$$

$$b. \begin{cases} 6x_1 - x_2 + 3x_3 = 4 \\ 9x_2 - x_3 = 1 \end{cases} \Rightarrow \begin{bmatrix} 6 & -1 & 3 & 4 \\ 0 & 5 & -1 & 1 \end{bmatrix}$$

$$c. \begin{cases} 2x_1 - 3x_4 + x_5 = 0 \\ -3x_1 - x_2 + x_3 = -1 \\ 6x_1 + 2x_2 - x_3 + 2x_4 - 3x_5 = 6 \end{cases} \Rightarrow \begin{bmatrix} 0 & 2 & 0 & -3 & 1 & 0 \\ -3 & -1 & 1 & 0 & 0 & -1 \\ 6 & 2 & -1 & 2 & -3 & 6 \end{bmatrix}$$

$$d. \{x_1 - x_3 = 7\} \Rightarrow [1 \ 0 \ 0 \ 0 \ -1 \ 7]$$

$$15. \begin{bmatrix} x_1^2 & x_1 & 1 & y_1 \\ x_2^2 & x_2 & 1 & y_2 \\ x_3^2 & x_3 & 1 & y_3 \end{bmatrix} = \begin{cases} a \cdot x_1^2 + b \cdot x_1 + c = y_1 \\ a \cdot x_2^2 + b \cdot x_2 + c = y_2 \\ a \cdot x_3^2 + b \cdot x_3 + c = y_3 \end{cases}$$

$$y = ax^2 + bx + c$$

$$(x_1, y_1), (x_2, y_2), (x_3, y_3)$$

$$17. \quad x_1 + Kx_2 = C$$

$$1 + Lx_2 = 3$$

$$Kx_2 = 3 - 1$$

$$Kx_2 = 2$$

$$x_2 = \frac{2}{K}$$

$$x_1 + Lx_2 = d$$

$$x_1 = d - Lx_2$$

$$1 + Lx_2 = 3$$

$$Lx_2 = 3 - 1$$

$$Lx_2 = 2$$

$$x_2 = \frac{2}{L}$$

$$K = L$$

$$C = d$$

$$(x_1, y)$$

$$x_1 = C - Kx_2$$

$$x_1 = d - Lx_2$$

$$\downarrow$$

$$C - Kx_2 = d - Lx_2$$

$$(L - K)x_2 = d - C$$

$$L - K = 0 \text{ e } d - C = 0$$

• Exercício 1.2 - Paros -

a. $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$ Escalonada. Coluna pivô não está vazia.

b. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$ Nenhum. Coluna 2 pivô não tem elementos a pivô e maior.

c. $\begin{bmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$ Escalonada. Coluna pivô com outros elementos.

d. $\begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$ Escalonada. Coluna pivô com elementos acima.

e. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow$ Nenhum. Linha 2 deveria estar no final.

f. $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 0 & 9 & 1 & 3 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow$ Nenhum. Linha 2 deveria ter o 1 à direita do pivô da linha 1.

g. $\begin{bmatrix} 1 & -7 & 0 & 1 \\ 0 & 0 & 1 & -7 \end{bmatrix} \rightarrow$ Escalonada e reduzida. Cumpre os 4 requisitos de escalonamento.

4. a. $\begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 7 \end{bmatrix} \rightarrow \begin{matrix} x = -3 \\ y = 0 \\ z = 7 \end{matrix} \quad S = (-3, 0, 7)$

b. $\begin{bmatrix} 1 & 0 & 0 & -7 & 9 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & 1 & -5 \end{bmatrix} \rightarrow \begin{matrix} a - 7d = 9 \\ b + 3d = 2 \\ c + d = -5 \end{matrix} \quad S = (8 + 7d, 2 - 3d, -5 - d, d, -)$

c. $\begin{bmatrix} 1 & -6 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{matrix} a - 6b + 3e = -2 \\ c + 4e = 7 \\ d + 5e = 8 \end{matrix} \quad S = (-2 + 6b + 3e, b, 7 - 4e, 8 - 5e, e)$

d. $\begin{bmatrix} 1 & -3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow SI$

$$6. \begin{cases} 2x_1 + 2x_2 + 2x_3 = 0 \\ -2x_1 + 5x_2 + 2x_3 = 1 \\ 8x_1 + x_2 + 4x_3 = -1 \end{cases} \Rightarrow \begin{bmatrix} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{bmatrix} \xrightarrow{\frac{1}{2}L_1} \begin{bmatrix} 1 & 1 & 1 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{bmatrix} \xrightarrow{2L_1+L_2} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 8 & 1 & 4 & -1 \end{bmatrix} \xrightarrow{-8L_1+L_3} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{bmatrix} \xrightarrow{\frac{1}{7}L_2} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & -7 & -4 & -1 \end{bmatrix} \xrightarrow{7L_2+L_3} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{L_1-L_2} \begin{bmatrix} 1 & 0 & \frac{3}{7} & -\frac{1}{7} \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 + \frac{3}{7}x_3 = -\frac{1}{7}$$

$$x_2 + \frac{4}{7}x_3 = \frac{1}{7}$$

$$x_3 = 3$$

$$S = \left(-\frac{1}{7} - \frac{3}{7} \cdot 3, \frac{1}{7} - \frac{4}{7} \cdot 3, 3\right)$$

SPI

$$8. \begin{cases} -2b + 3c = 1 \\ 3a + 6b - 3c = -2 \\ 6a + 6b + 3c = 5 \end{cases} \Rightarrow \begin{bmatrix} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{bmatrix} \xrightarrow{L_1 \leftrightarrow L_2} \begin{bmatrix} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{bmatrix} \xrightarrow{\frac{1}{3}L_1} \begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{bmatrix} \xrightarrow{L_3 - 6L_1} \begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 0 & 6 & 9 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 0 & 6 & 9 & 9 \end{bmatrix} \xrightarrow{-\frac{1}{2}L_2} \begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 6 & 9 & 9 \end{bmatrix} \xrightarrow{L_3 + 6L_2} \begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & 6 \end{bmatrix} \quad 0a + 0b + 0c = 6$$

SI

$$10. \begin{cases} 2x_1 + 2x_2 + 2x_3 = 0 \\ -2x_1 + 5x_2 + 2x_3 = 1 \\ 8x_1 + x_2 + 4x_3 = -1 \end{cases} \Rightarrow \begin{bmatrix} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{bmatrix} \xrightarrow{\frac{1}{2}L_1} \begin{bmatrix} 1 & 1 & 1 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{bmatrix} \xrightarrow{2L_1+L_2} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 8 & 1 & 4 & -1 \end{bmatrix} \xrightarrow{-8L_1+L_3} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{bmatrix} \xrightarrow{\frac{1}{7}L_2} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & -7 & -4 & -1 \end{bmatrix} \xrightarrow{7L_2+L_3} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} a+b+c &= 0 \\ b+\frac{4}{7}c &= \frac{1}{7} \end{aligned} \quad S = \left(-b-c, \frac{1}{7} - \frac{4}{7}c, c\right)$$

$$12. \begin{bmatrix} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{bmatrix} \xrightarrow{L_1 \leftrightarrow L_2} \begin{bmatrix} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{bmatrix} \xrightarrow{\frac{1}{3}L_1} \begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{bmatrix} \xrightarrow{L_3 - 6L_1} \begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 0 & 6 & 9 & 9 \end{bmatrix} \xrightarrow{\frac{1}{2}L_2} \begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 6 & 9 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 6 & 9 & 9 \end{bmatrix} \xrightarrow{L_3 + 6L_2} \begin{bmatrix} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & 6 \end{bmatrix} \quad \begin{cases} a+2b-c = -\frac{2}{3} \\ b-\frac{3}{2}c = -\frac{1}{2} \\ 0a+0b+0c = 6 \end{cases} \quad SI$$

$$14. \begin{cases} x_1 + 3x_2 - x_3 = 0 \\ x_2 - 8x_3 = 0 \\ 4x_3 = 0 \end{cases} \Rightarrow \begin{bmatrix} 1 & 3 & -1 & 0 \\ 0 & 1 & -8 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} \quad \text{Redu\c{c}\~ao n\~ao-trivial}$$

$$16. \begin{cases} 3x_1 - 2x_2 = 0 \\ 6x_1 - 4x_2 = 0 \end{cases} \quad \text{Redu\c{c}\~ao Trivial}$$

$$18. \begin{cases} 2x - y - 3z = 0 \\ -x + 2y + 3z = 0 \\ x + y + 4z = 0 \end{cases} \Rightarrow \begin{bmatrix} 2 & -1 & -3 & 0 \\ -1 & 2 & 3 & 0 \\ 1 & 1 & 4 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}L_1} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} & 0 \\ -1 & 2 & 3 & 0 \\ 1 & 1 & 4 & 0 \end{bmatrix} \xrightarrow{L_3+L_1} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & \frac{3}{2} & -\frac{9}{2} & 0 \\ 1 & 1 & 4 & 0 \end{bmatrix} \xrightarrow{L_3-L_1} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & \frac{3}{2} & -\frac{9}{2} & 0 \\ 0 & \frac{3}{2} & \frac{11}{2} & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & \frac{3}{2} & -\frac{9}{2} & 0 \\ 0 & \frac{3}{2} & \frac{11}{2} & 0 \end{bmatrix} \xrightarrow{-\frac{2}{3}L_2} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & 1 & -3 & 0 \\ 0 & \frac{3}{2} & \frac{11}{2} & 0 \end{bmatrix} \xrightarrow{L_3-\frac{3}{2}L_2} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & \frac{9}{2} & 0 \end{bmatrix} \xrightarrow{L_1+\frac{1}{2}L_2} \begin{bmatrix} 1 & 0 & -3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & \frac{9}{2} & 0 \end{bmatrix}$$

$$\xrightarrow{L_3+\frac{2}{9}L_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{\frac{2}{9}L_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad \begin{matrix} x=0 \\ y=0 \\ z=0 \end{matrix} \quad \text{Redu\c{c}\~ao Trivial}$$

$$20. \begin{cases} u + 3w - 2x = 0 \\ 2u + w - 4v + 3x = 0 \\ 2u + 3w + 2v - x = 0 \\ -4u - 3w + 5v - 4x = 0 \end{cases} \Rightarrow \begin{bmatrix} 0 & 1 & 3 & -2 & 0 \\ 2 & 1 & -4 & 3 & 0 \\ 2 & 3 & 2 & -1 & 0 \\ -4 & -3 & 5 & -4 & 0 \end{bmatrix} \xrightarrow{\substack{L_2-L_1 \\ L_3-L_1}} \begin{bmatrix} 2 & 1 & -4 & 3 & 0 \\ 0 & 1 & 3 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 \\ -4 & -3 & 5 & -4 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}L_1} \begin{bmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & 3 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 \\ -4 & -3 & 5 & -4 & 0 \end{bmatrix}$$

$$\xrightarrow{L_3-2L_1} \begin{bmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & 3 & -2 & 0 \\ 0 & 2 & -1 & -4 & 0 \\ -4 & -3 & 5 & -4 & 0 \end{bmatrix} \xrightarrow{L_4+4L_1} \begin{bmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & 3 & -2 & 0 \\ 0 & 2 & -1 & -4 & 0 \\ 0 & -1 & -3 & 8 & 0 \end{bmatrix} \xrightarrow{L_3-2L_2} \begin{bmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & 3 & -2 & 0 \\ 0 & 0 & -7 & 12 & 0 \\ 0 & -1 & -3 & 8 & 0 \end{bmatrix} \xrightarrow{L_4+L_2} \begin{bmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & 3 & -2 & 0 \\ 0 & 0 & -7 & 12 & 0 \\ 0 & 0 & 1 & 6 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & 3 & -2 & 0 \\ 0 & 0 & -7 & 12 & 0 \\ 0 & 0 & 1 & 6 & 0 \end{bmatrix} \xrightarrow{-\frac{1}{7}L_3} \begin{bmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & 3 & -2 & 0 \\ 0 & 0 & 1 & -\frac{12}{7} & 0 \\ 0 & 0 & 1 & 6 & 0 \end{bmatrix} = \begin{cases} u + \frac{1}{2}w - 2x = 0 \\ w - 3x = 0 \\ w - \frac{12}{7}x = 0 \\ 6x = 0 \end{cases} \quad \text{SPT}$$

$$22. \begin{cases} x_1 + 3x_2 + x_4 = 0 \\ x_1 + 4x_2 + 2x_3 = 0 \\ -2x_2 - 2x_3 - x_4 = 0 \\ 2x_1 - 4x_2 + x_3 + x_4 = 0 \\ x_1 - 2x_2 - x_3 + x_4 = 0 \end{cases} \Rightarrow \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 1 & 4 & 2 & 0 & 0 \\ 0 & -2 & -2 & -1 & 0 \\ 2 & -4 & 1 & 1 & 0 \\ 1 & -2 & -1 & 1 & 0 \end{bmatrix} \xrightarrow{L_2-L_1, L_3-L_1, L_4-2L_1, L_5-L_1} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & -2 & -2 & -1 & 0 \\ 2 & -4 & 1 & 1 & 0 \\ 1 & -2 & -1 & 1 & 0 \end{bmatrix} \xrightarrow{L_4-2L_1, L_5-L_1} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & -2 & -2 & -1 & 0 \\ 0 & -8 & 1 & -1 & 0 \\ 0 & -5 & -1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & -2 & -2 & -1 & 0 \\ 0 & -8 & 1 & -1 & 0 \\ 0 & -5 & -1 & 0 & 0 \end{bmatrix} \xrightarrow{L_5-L_1} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & -2 & -2 & -1 & 0 \\ 0 & -8 & 1 & -1 & 0 \\ 0 & -5 & -1 & 0 & 0 \end{bmatrix} \xrightarrow{L_3+2L_2, L_4+8L_2, L_5+5L_2} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 4 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 9 & -1 & 0 \end{bmatrix} \xrightarrow{L_4+10L_3, L_5-9L_3} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 4 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -9 & 0 \end{bmatrix}$$

$$\xrightarrow{L_5-L_2} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 4 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & -3 & 0 & 0 \end{bmatrix} \xrightarrow{L_3+L_5, L_4+9L_5} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & -3 & 0 & 0 \end{bmatrix} \xrightarrow{L_4-L_3, L_5+3L_3} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 3 & 0 \end{bmatrix}$$

$$\xrightarrow{-\frac{1}{6}L_4} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 & 0 \end{bmatrix} \xrightarrow{L_5-3L_4} \begin{bmatrix} 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 + 3x_2 + x_4 = 0 \\ x_2 + 2x_3 = 0 \\ x_3 + x_4 = 0 \\ x_4 = 0 \\ 0x_4 = 0 \end{cases} \quad \text{SPI}$$

$$24. \begin{cases} z_3 + z_4 + z_5 = 0 \\ -z_1 - z_2 + 2z_3 - 3z_4 + z_5 = 0 \\ z_1 + z_2 - 2z_3 - z_5 = 0 \\ 2z_1 + 2z_2 - z_3 + z_5 = 0 \end{cases} \Rightarrow \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 2 & 2 & -1 & 0 & 1 \end{bmatrix} \xrightarrow{L_2+L_1, L_3-L_1, L_4-2L_1} \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 2 & 2 & -1 & 0 & 1 \end{bmatrix} \xrightarrow{L_2+L_1, L_3-L_1, L_4-2L_1} \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 2 & 2 & -1 & 0 & 1 \end{bmatrix} \xrightarrow{L_2+L_1, L_3-L_1, L_4-2L_1} \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 2 & 2 & -1 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{L_5-2L_1} \begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -4 & 0 & 3 \end{bmatrix} \xrightarrow{L_2+3L_3, L_4+4L_3} \begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -3 & 1 \end{bmatrix} \xrightarrow{L_2-L_3, L_4-L_3} \begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -3 & 1 \end{bmatrix} \xrightarrow{L_2+L_4} \begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -3 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} z_1 + z_2 - 2z_3 - z_5 = 0 \\ z_3 + z_4 + z_5 = 0 \\ z_4 + 3z_5 = 0 \\ -3z_4 + z_5 = 0 \end{cases} \quad \text{SPI}$$

$$26. \begin{cases} x + 2y + z = 2 \\ 2x - 2y + 3z = 1 \\ x + 2y - (a^2 - 3)z = a \end{cases} \Rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & -2 & 3 & 1 \\ 1 & 2 & -(a^2-3)a \end{bmatrix} \xrightarrow{L_2-2L_1} \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -6 & 1 & -3 \\ 1 & 2 & -(a^2-3)a \end{bmatrix} \xrightarrow{L_3-L_1} \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -6 & 1 & -3 \\ 0 & 0 & -a^2+2a-2 \end{bmatrix}$$

$$-a^2+2 = a-2 \quad 1 \text{ solução (SPD)} \quad -a^2+2=0 \Rightarrow -a=\pm\sqrt{2} \quad a \neq \pm\sqrt{2}$$

Inf. soluções (SPI) - Não existe nesse sistema.

$$SI \quad a = \pm\sqrt{2}$$

$$28. \begin{cases} x + y + 7z = -7 \\ 2x + 3y + 17z = -16 \\ x + 2y + (a^2+1)z = 3a \end{cases} \Rightarrow \begin{bmatrix} 1 & 1 & 7 & -7 \\ 2 & 3 & 17 & -16 \\ 1 & 2 & (a^2+1) & 3a \end{bmatrix} \xrightarrow{L_2-2L_1} \begin{bmatrix} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 1 & 2 & (a^2+1) & 3a \end{bmatrix} \xrightarrow{L_3-L_1} \begin{bmatrix} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 1 & (a^2-6) & 3a+7 \end{bmatrix} \xrightarrow{L_3-L_2} \begin{bmatrix} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & (a^2-9) & 3a+9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & (a^2-9) & 3a+9 \end{bmatrix} \Rightarrow \begin{cases} a^2-9 = 3a+9 \\ a^2-9 = 0 \\ a = \pm 3 \\ 0 = 3 \end{cases}$$

$$\begin{cases} a = 3 \\ a^2-9 = 3a+9 \\ 0 = 18 \end{cases}$$

$$\begin{cases} a = -3 \\ (-3)^2-9 = 3(-3)+9 \\ 0 = 0 \end{cases}$$

$$SI \quad a = 3$$

$$\text{Inf. Sol. (SPI)} \quad a = -3$$

$$1 \text{ solução (SPD)} \quad a \neq \pm 3$$

$$30. \begin{cases} x_1 + x_2 + x_3 = a \\ 2x_1 + 2x_3 = b \\ 3x_2 + 3x_3 = c \end{cases} \Rightarrow \begin{bmatrix} 1 & 1 & 1 & a \\ 2 & 0 & 2 & b \\ 0 & 3 & 3 & c \end{bmatrix} \xrightarrow{L_2-2L_1} \begin{bmatrix} 1 & 1 & 1 & a \\ 0 & -2 & 0 & b-2a \\ 0 & 3 & 3 & c \end{bmatrix} \xrightarrow{\frac{1}{2}L_2} \begin{bmatrix} 1 & 1 & 1 & a \\ 0 & 1 & 0 & \frac{b-2a}{2} \\ 0 & 3 & 3 & c \end{bmatrix}$$

$$\xrightarrow{L_3-3L_2} \begin{bmatrix} 1 & 1 & 1 & a \\ 0 & 1 & 0 & \frac{b-2a}{2} \\ 0 & 0 & 3 & c - \frac{3}{2}(b-2a) \end{bmatrix} \Rightarrow 3x_3 = c - 3a + \frac{3b}{2}$$

Sistema consistente para qualquer real de a, b, c .

$$32. \begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & -29 \\ 3 & 4 & 5 \end{bmatrix} \xrightarrow{2L_3 - 3L_1} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & -29 \\ 0 & 5 & 1 \end{bmatrix} \xrightarrow{2L_3 + 5L_2} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & -29 \\ 0 & 0 & -143 \end{bmatrix} \xrightarrow{(L_3) \cdot (-29)L_2} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -296 & 0 \\ 0 & 0 & -143 \end{bmatrix} \xrightarrow{2L_1 + L_2} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -296 & 0 \\ 0 & 0 & -143 \end{bmatrix}$$

$$\begin{bmatrix} 577 & 0 & 0 \\ 0 & -296 & 0 \\ 0 & 0 & -143 \end{bmatrix} \quad \begin{aligned} L_1 &\leftarrow L_1 / 577 \\ L_2 &\leftarrow L_2 / -296 \\ L_3 &\leftarrow L_3 / -143 \end{aligned}$$

$$34. \begin{cases} 2 \sin \alpha - \cos \beta + 3 \tan \gamma = 3 \\ 4 \sin \alpha + 2 \cos \beta - 2 \tan \gamma = 2 \\ 5 \sin \alpha - 3 \cos \beta + \tan \gamma = 9 \end{cases} = \begin{bmatrix} 2 & -1 & 3 & 3 \\ 4 & 2 & -2 & 2 \\ 5 & -3 & 1 & 9 \end{bmatrix} \xrightarrow{\substack{L_2 - 2L_1 \\ L_3 - 3L_1}} \begin{bmatrix} 2 & -1 & 3 & 3 \\ 0 & 4 & -8 & -4 \\ 0 & 0 & -9 & 0 \end{bmatrix}$$

$$\begin{aligned} -8z &= 0 & 4y - 8z &= -4 & 2x - (-1) + 3(0) &= 3 & \sin \alpha &= \frac{\pi}{2} \\ z &= 0 & 4y &= -4 & 2x &= 2 & \cos \alpha &= \frac{\pi}{2} \\ 0 &= 0 & y &= -1 & x &= 1 & \tan \alpha &= 0 \end{aligned} \quad S = \left(\frac{\pi}{2}, \frac{\pi}{2}, 0 \right)$$

$$36. \begin{cases} \frac{1}{x} + \frac{2}{y} - \frac{4}{z} = 1 \\ \frac{2}{x} + \frac{3}{y} + \frac{8}{z} = 0 \\ -\frac{1}{x} + \frac{9}{y} + \frac{10}{z} = 5 \end{cases} = \begin{bmatrix} \frac{1}{x} & \frac{2}{y} & -\frac{4}{z} & 1 \\ \frac{2}{x} & \frac{3}{y} & \frac{8}{z} & 0 \\ -\frac{1}{x} & \frac{9}{y} & \frac{10}{z} & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -4 & 1 \\ 2 & 3 & 8 & 0 \\ -1 & 9 & 10 & 5 \end{bmatrix} \xrightarrow{\substack{L_2 - 2L_1 \\ L_3 + L_1}}$$

$$a = \frac{1}{x} \quad b = \frac{1}{y} \quad c = \frac{1}{z}$$

$$\begin{bmatrix} 1 & 2 & -4 & 1 \\ 0 & -1 & 16 & -2 \\ 0 & 11 & 6 & 6 \end{bmatrix} \xrightarrow{L_3 + 11L_2} \begin{bmatrix} 1 & 2 & -4 & 1 \\ 0 & -1 & 16 & -2 \\ 0 & 0 & 182 & -16 \end{bmatrix}$$

$$\begin{aligned} 182c &= -16 & -b + 16c &= -2 & a + 2b - 4c &= 1 \\ c &= -\frac{8}{91} & b &= \frac{54}{91} & a &= -\frac{49}{91} \end{aligned}$$

$$x = \frac{1}{a} = -\frac{91}{49} \quad y = \frac{1}{b} = \frac{91}{54} \quad z = \frac{1}{c} = -\frac{91}{8}$$

$$S = \left(-\frac{13}{7}, \frac{91}{54}, -\frac{91}{8} \right)$$

$$38. ax^2 + ay^2 + bx + cy + d = 0$$

$$(-2, 7), (-4, 5), (4, -3)$$

$$a(-2)^2 + a(7)^2 + b(-2) + c(7) + d = 0$$

$$53a - 2b + 7c + d = 0$$

$$a(-4)^2 + a(5)^2 + b(-4) + c(5) + d = 0$$

$$41a - 4b + 5c + d = 0$$

$$a(4)^2 + a(-3)^2 + b(4) + c(-3) + d = 0$$

$$25a + 4b - 3c + d = 0$$

$$a = 1$$

$$\begin{cases} -2b + 7c + d = -53 \\ -4b + 5c + d = -41 \\ 4b - 3c + d = -25 \end{cases} = \begin{bmatrix} -2 & 7 & 1 & -53 \\ -4 & 5 & 1 & -41 \\ 4 & -3 & 1 & -25 \end{bmatrix} \xrightarrow{\substack{L_2 + 2L_1 \\ L_3 + 2L_1}} \begin{bmatrix} -2 & 7 & 1 & -53 \\ 0 & -9 & -1 & 65 \\ 0 & 11 & 3 & -131 \end{bmatrix} \xrightarrow{9L_3 + 11L_2} \begin{bmatrix} -2 & 7 & 1 & -53 \\ 0 & -9 & -1 & 65 \\ 0 & 0 & 16 & -464 \end{bmatrix}$$

$$16d = -464 \quad -9c - (-29) = 65 \quad -2b + 7(-4) + (-29) = -53$$

$$d = -29$$

$$c = -4$$

$$b = -2$$

$$S = \{1, -2, -4, -29\}$$

40. a. $\begin{pmatrix} a & b & c & d & e \\ a & b & c & d & e \\ a & b & c & d & e \end{pmatrix}$ Número máximo = 3 pivôs.

b. $B_{3 \times 6} = \begin{pmatrix} a & b & c & d & e & 0 \\ a & b & c & d & e & 0 \\ a & b & c & d & e & 0 \end{pmatrix}$ Num. - Num. = $5 - 1 = 4$
 De a matriz for não-nula = 4 pivôs.

c. $C_{5 \times 3} =$ N° máximo de pivôs = 3. No mínimo 2 linhas de zeros.
 $5 - 3 = 2$

42. $\begin{cases} ax + by = 0 \\ cx + dy = 0 \\ ex + fy = 0 \end{cases}$ a. Se a única solução for trivial $(0, 0)$, não existe outro ponto em comum além da origem (distintos e concorrentes na origem).

b. Se o sistema tiver solução não-trivial $(x, y \neq 0)$, as três retas são coincidentes.

Exercício 1.3 - 1, 3, 5, 7, 9, 11 -

1-a. $A \quad B \quad C \quad D \quad E$
 $(4 \times 5) \quad (4 \times 5) \quad (5 \times 2) \quad (4 \times 2) \quad (5 \times 4)$

$B.A =$ Não definida

$4 \times 5, 4 \times 5 =$ Não é possível $\times$

b. $AC + D = A.C = 4 \times 2$
 $AC + D = 4 \times 2 + 4 \times 2 = 4 \times 2$
 $\checkmark$

c. $AE + B = AC = 4 \times 4$
 $AE + B = 4 \times 4 + 4 \times 5 \rightarrow$ Não definida $\times$

d. $AB + D = AD =$ Não definida $\times$

e. $C(A+B) = CA = 5 \times 5$
 $CB = 5 \times 5$
 $CA + CB = 5 \times 5 + 5 \times 5 = 5 \times 5$
 $\checkmark$

f. $C(AC) = A.C = 4 \times 5, 5 \times 2 = 4 \times 2$
 $CA.C = 5 \times 4, 4 \times 2 = 5 \times 2$
 $\checkmark$

g. $E^T A = E^T = 4 \times 5$
 $E^T A = 4 \times 5, 4 \times 5 =$ Não definida $\times$

h. $(A^T + E)D = A^T = 5 \times 4$
 $A^T + E = 5 \times 4 + 5 \times 4 = 5 \times 4$
 $\cdot D = 5 \times 4, 4 \times 2 = 5 \times 2$
 $\checkmark$

3-a. $D + E = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 7 & 6 & 5 \\ -2 & 1 & 3 \\ 7 & 3 & 7 \end{bmatrix}$

b. $D - E = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} - \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -5 & 4 & -1 \\ 0 & -1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$

$$c. 5A = 5 \cdot \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 15 & 0 \\ -5 & 10 \\ 5 & 5 \end{bmatrix}$$

$$d. -7C = -7 \begin{bmatrix} 1 & 4 & 0 \\ 3 & 1 & 5 \end{bmatrix} = \begin{bmatrix} -7 & -28 & -14 \\ -21 & -7 & -35 \end{bmatrix}$$

$$e. 2B - C = 2 \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \rightarrow \text{Não é possível}$$

$$f. 4E - 2D = 2 \cdot \begin{bmatrix} 1 & 5 & 0 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} + 4 \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 0 \\ 4 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & -10 & -4 \\ 2 & 0 & -2 \\ -6 & -4 & -8 \end{bmatrix} + \begin{bmatrix} 24 & 4 & 12 \\ -4 & 4 & 8 \\ 16 & 4 & 12 \end{bmatrix} = \begin{bmatrix} 22 & -6 & 8 \\ -2 & 4 & 6 \\ 10 & 0 & 4 \end{bmatrix}$$

$$g. -3(D+2C) = -3 \cdot \left(\begin{bmatrix} 1 & 5 & 0 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 12 & 2 & 6 \\ -2 & 3 & 6 \\ 8 & 2 & 6 \end{bmatrix} \right) = -3 \cdot \begin{bmatrix} 13 & 7 & 6 \\ -3 & 3 & 7 \\ 11 & 4 & 10 \end{bmatrix} = \begin{bmatrix} -39 & -21 & -18 \\ 9 & -9 & -21 \\ -33 & -12 & -30 \end{bmatrix}$$

$$h. A - A = 0$$

$$i. \ln(D) = 1 + 0 + 4 = 5$$

$$j. \ln(D \cdot 3C) = 5 + 3 \cdot 10 = 35$$

$$\ln(C) = 6 + 1 + 3 = 10$$

$$k. 4 \ln(7B) = 4 \cdot (7 \cdot 6) = 4 \cdot 42 = 168$$

$$\ln(D) = 4 + 2 = 6$$

$$l. \ln(A) = \text{Não é possível}$$

$$5a. AB = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}_{3 \times 2} \cdot \begin{bmatrix} 4 & -1 \\ 0 & 0 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 3 \cdot 4 + 0 \cdot 0 & 3 \cdot (-1) + 0 \cdot 0 \\ -1 \cdot 4 + 2 \cdot 0 & -1 \cdot (-1) + 2 \cdot 0 \\ 1 \cdot 4 + 1 \cdot 0 & 1 \cdot (-1) + 1 \cdot 0 \end{bmatrix}_{3 \times 2} = \begin{bmatrix} 12 & -3 \\ -4 & 2 \\ 4 & -1 \end{bmatrix}$$

$$b. B \cdot A = \text{Não é possível } (2 \times 2, 3 \times 2).$$

$$c. (3E)D = 3E \cdot \begin{bmatrix} 18 & 3 & 9 \\ -3 & 3 & 6 \\ 12 & 3 & 9 \end{bmatrix} \cdot \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 18 \cdot 1 + 3 \cdot (-1) + 9 \cdot 3 & 18 \cdot 5 + 3 \cdot 0 + 9 \cdot 2 & 18 \cdot 2 + 3 \cdot 1 + 9 \cdot 4 \\ -3 \cdot 1 + 3 \cdot (-1) + 6 \cdot 3 & -3 \cdot 5 + 3 \cdot 0 + 6 \cdot 2 & -3 \cdot 2 + 3 \cdot 1 + 6 \cdot 4 \\ 12 \cdot 1 + 3 \cdot (-1) + 9 \cdot 3 & 12 \cdot 5 + 3 \cdot 0 + 9 \cdot 2 & 12 \cdot 2 + 3 \cdot 1 + 9 \cdot 4 \end{bmatrix} = \begin{bmatrix} 42 & 108 & 75 \\ 12 & -3 & 21 \\ 36 & 78 & 63 \end{bmatrix}$$

$$d. (AB) \cdot C = \begin{bmatrix} 12 & -3 \\ -4 & 2 \\ 4 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 12 \cdot 1 + (-3) \cdot 3 & 12 \cdot 4 + (-3) \cdot 1 & 12 \cdot 2 + (-3) \cdot 5 \\ -4 \cdot 1 + 2 \cdot 3 & -4 \cdot 4 + 2 \cdot 1 & -4 \cdot 2 + 2 \cdot 5 \\ 4 \cdot 1 + (-1) \cdot 3 & 4 \cdot 4 + (-1) \cdot 1 & 4 \cdot 2 + (-1) \cdot 5 \end{bmatrix} = \begin{bmatrix} 3 & 45 & 9 \\ 11 & -11 & 17 \\ 7 & 17 & 13 \end{bmatrix}$$

$$e. A(BC) = \begin{bmatrix} 3 & 45 & 9 \\ 11 & -11 & 17 \\ 7 & 17 & 13 \end{bmatrix} \rightarrow \text{Propriedade associativa (Mesmo da letra d)}.$$

$$f. CCT: \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 4 \cdot 4 + 2 \cdot 2 & 1 \cdot 3 + 4 \cdot 1 + 2 \cdot 5 \\ 3 \cdot 1 + 1 \cdot 4 + 5 \cdot 2 & 3 \cdot 3 + 1 \cdot 1 + 5 \cdot 5 \end{bmatrix} = \begin{bmatrix} 21 & 17 \\ 17 & 35 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix}$$

$$g. (DA)^T = DA = \begin{bmatrix} 0 & 12 \\ -2 & 1 \\ 11 & 8 \end{bmatrix} \quad DA^T = \begin{bmatrix} 0 & -2 & 11 \\ 12 & 1 & 8 \end{bmatrix}$$

$$h. (C^T B) \cdot A^T = C^T B = \begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix} \cdot \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 16 & -2 \\ 8 & 9 \end{bmatrix} \quad \begin{bmatrix} 4 & 5 \\ 16 & -2 \\ 8 & 9 \end{bmatrix} \cdot \begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 12 & 6 & 9 \\ 48 & -20 & 14 \\ 24 & 8 & 16 \end{bmatrix}$$

$$i. \ln D \cdot D^T = 5 \cdot 5 = 25$$

$$\ln D = 1 + 0 + 4 = 5$$

$$\ln D^T = 5$$

$$D^T = \begin{bmatrix} 1 & -1 & 3 \\ 5 & 0 & 2 \\ 2 & 1 & 4 \end{bmatrix}$$

$$j = \ln(4e^T \cdot D) = 4 \cdot 10 \cdot 5 = 55$$

$$\ln(E) = 6 + 1 + 3 = 10$$

$$\ln(D) = 1 + 0 + 4 = 5$$

$$K = \ln(C^T A^T + 2E^T) \quad AC = \begin{bmatrix} 3 & 12 & 6 \\ 5 & -2 & 8 \\ 4 & 3 & 7 \end{bmatrix}$$

$$\ln(C^T A^T + 2E^T) = 20 + 8 = 28$$

$$T_n(C^T A^T) = 3 \cdot 2 + 7 = 8$$

$$\ln(2E^T) = 2 \cdot 10 = 20$$

$$2. \ln((CC^T)^T A)$$

$$(CC^T)^T A = (CC^T) A = C(C^T A)$$

$$C^T A = \begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 23 & 2 \\ 3 & 3 \\ 10 & 7 \end{bmatrix}$$

$$C(C^T A) = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \begin{bmatrix} 23 & 2 \\ 3 & 3 \\ 10 & 7 \end{bmatrix} = \begin{bmatrix} 23 + 12 + 20 & 2 + 12 + 14 \\ 69 + 13 + 50 & 6 + 3 + 35 \end{bmatrix} = \begin{bmatrix} 55 & 29 \\ 132 & 44 \end{bmatrix}$$

$$\ln = 55 + 44 = 99$$

$$3-a. \begin{bmatrix} 3 & -2 & 7 \end{bmatrix} \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix} \quad \begin{aligned} C_{11} &= 3 \cdot 6 + (-2) \cdot 0 + 7 \cdot 7 = 67 \\ C_{12} &= 3 \cdot (-2) + (-2) \cdot 1 + 7 \cdot 7 = 41 \\ C_{13} &= 3 \cdot 4 + (-2) \cdot 3 + 7 \cdot 5 = 41 \end{aligned}$$

$$\begin{bmatrix} 67 & 41 & 41 \end{bmatrix}$$

$$b. \begin{bmatrix} 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix} \quad \begin{aligned} C_{31} &= 0 \cdot 6 + 4 \cdot 0 + 9 \cdot 7 = 63 \\ C_{32} &= 0 \cdot (-2) + 4 \cdot 1 + 9 \cdot 7 = 67 \\ C_{33} &= 0 \cdot 4 + 4 \cdot 3 + 9 \cdot 5 = 57 \end{aligned}$$

$$\begin{bmatrix} 63 & 67 & 57 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix} \quad \begin{aligned} C_{11} &= 3 \cdot (-2) + (-2) \cdot 1 + 7 \cdot 7 = 41 \\ C_{21} &= 6 \cdot (-2) + 5 \cdot 1 + 4 \cdot 7 = 31 \\ C_{31} &= 0 \cdot (-2) + 4 \cdot 1 + 9 \cdot 7 = 67 \end{aligned} \quad \begin{bmatrix} 41 \\ 31 \\ 67 \end{bmatrix}$$

$$d. \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \quad \begin{aligned} C_{11} &= 6 \cdot 3 - 2 \cdot 6 + 4 \cdot 0 = 6 \\ C_{21} &= 0 \cdot 3 + 1 \cdot 6 + 3 \cdot 0 = 6 \\ C_{31} &= 7 \cdot 3 + 7 \cdot 6 + 5 \cdot 0 = 63 \end{aligned} \quad \begin{bmatrix} 6 \\ 6 \\ 63 \end{bmatrix}$$

$$e. \begin{bmatrix} 0 & 4 & 9 \\ 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \quad \begin{aligned} C_{11} &= 0 \cdot 3 + 4 \cdot 6 + 9 \cdot 0 = 24 \\ C_{21} &= 0 \cdot 3 + 4 \cdot 6 + 9 \cdot 0 = 24 \\ C_{31} &= 0 \cdot 3 + 4 \cdot 6 + 9 \cdot 0 = 24 \\ C_{41} &= 0 \cdot 3 + 4 \cdot 6 + 9 \cdot 0 = 24 \end{aligned} \quad \begin{bmatrix} 24 & 56 & 37 \end{bmatrix}$$

$$f. \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \\ 9 \end{bmatrix} \quad \begin{aligned} C_{13} &= 3 \cdot 7 - 2 \cdot 4 + 7 \cdot 9 = 76 \\ C_{23} &= 6 \cdot 7 + 5 \cdot 4 + 4 \cdot 9 = 98 \\ C_{33} &= 0 \cdot 7 + 4 \cdot 4 + 9 \cdot 9 = 97 \end{aligned} \quad \begin{bmatrix} 76 \\ 98 \\ 97 \end{bmatrix}$$

$$g. a. \text{ Column } 1 = 3a_1 + 6a_2 + 0a_3 \quad (3, 6, 0) \\ \text{Column } 2 = -2a_1 + 5a_2 + 4a_3 \quad (-2, 5, 4) \\ \text{Column } 3 = 7a_1 + 4a_2 + 9a_3 \quad (7, 4, 9)$$

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AA $\rightarrow a_1, a_2, a_3$

$$b. \text{ Column } 1 = 6b_1 + 0b_2 + 7b_3 \quad (6, 0, 7) \\ \text{Column } 2 = -2b_1 + 1b_2 + 7b_3 \quad (-2, 1, 7) \\ \text{Column } 3 = 4b_1 + 3b_2 + 5b_3 \quad (4, 3, 5)$$

$$11. a. A = \begin{bmatrix} 2 & -3 & 5 \\ 9 & -1 & 1 \\ 1 & 5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \\ 0 \end{bmatrix}$$

$$b. A = \begin{bmatrix} 4 & 0 & -3 & 1 \\ 5 & 1 & 0 & -8 \\ 2 & -5 & 9 & -1 \\ 0 & 3 & -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}$$

Exercício 1.4 - (1, 3, 5, 7, 9, 11).

1. a. $A + (B + C)$

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ -2 & 1 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 8 & -3 & -5 \\ 0 & 1 & 2 \\ 4 & -7 & 6 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & -2 & 3 \\ 1 & 7 & 4 \\ 3 & 5 & 9 \end{bmatrix}$$

$$B + C = \begin{bmatrix} 8 & -5 & -2 \\ 1 & 8 & 6 \\ 7 & -2 & 15 \end{bmatrix}$$

$$A + (B + C) = \begin{bmatrix} 10 & -6 & 1 \\ 1 & 12 & 11 \\ 5 & -1 & 19 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 10 & -4 & -2 \\ 0 & 5 & 7 \\ 2 & -6 & 10 \end{bmatrix}$$

$$(A + B) + C = \begin{bmatrix} 10 & -6 & 1 \\ 1 & 12 & 11 \\ 5 & -1 & 19 \end{bmatrix}$$

b. $(AB)C = A(BC)$

$$AB = \begin{bmatrix} 2 \cdot 8 + (-1) \cdot 0 + 3 \cdot 4 & 2 \cdot (-3) + (-1) \cdot 1 + 3 \cdot 7 & 2 \cdot (-5) + (-1) \cdot 2 + 3 \cdot 6 \\ 0 \cdot 8 + 4 \cdot 0 + 5 \cdot 4 & 0 \cdot (-3) + 4 \cdot 1 + 5 \cdot 7 & 0 \cdot (-5) + 4 \cdot 2 + 5 \cdot 6 \\ -2 \cdot 8 + 1 \cdot 0 + 4 \cdot 4 & -2 \cdot (-3) + 1 \cdot 1 + 4 \cdot 7 & -2 \cdot (-5) + 1 \cdot 2 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 22 & -28 & 6 \\ 20 & -31 & 32 \\ 0 & -21 & 36 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} 25 \cdot 0 + (-25) \cdot 1 + 6 \cdot 3 & 25 \cdot (-2) + (-25) \cdot 7 + 6 \cdot 5 & 25 \cdot 3 + (-25) \cdot 4 + 6 \cdot 9 \\ 90 \cdot 0 + (-31) \cdot 1 + 38 \cdot 3 & 90 \cdot (-2) + (-31) \cdot 7 + 38 \cdot 5 & 90 \cdot 3 + (-31) \cdot 4 + 38 \cdot 9 \\ 0 \cdot 0 + (-21) \cdot 1 + 36 \cdot 3 & 0 \cdot (-2) + (-21) \cdot 7 + 36 \cdot 5 & 0 \cdot 3 + (-21) \cdot 4 + 36 \cdot 9 \end{bmatrix} = \begin{bmatrix} -10 & -222 & 22 \\ 83 & -67 & 278 \\ 87 & 33 & 240 \end{bmatrix}$$

$$BC = \begin{bmatrix} 8 & -3 & -5 \\ 0 & 1 & 2 \\ 4 & -7 & 6 \end{bmatrix} \cdot \begin{bmatrix} 0 & -2 & 3 \\ 1 & 7 & 4 \\ 3 & 5 & 9 \end{bmatrix} = \begin{bmatrix} -18 & -62 & -33 \\ 7 & 17 & 22 \\ 11 & -27 & 38 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ -2 & 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} -18 & -62 & -33 \\ 7 & 17 & 22 \\ 11 & -27 & 38 \end{bmatrix} = \begin{bmatrix} -10 & -222 & 22 \\ 83 & -67 & 278 \\ 87 & 33 & 240 \end{bmatrix}$$

c. $(a+b)C = aC + bC$

$$a = 4$$

$$b = -7$$

$$a + b = 4 - 7 = -3$$

$$-3C = \begin{bmatrix} 0 & 6 & -9 \\ -3 & -21 & -12 \\ -9 & -15 & -27 \end{bmatrix}$$

$$aC = \begin{bmatrix} 0 & -8 & 12 \\ 4 & 28 & 16 \\ 12 & 20 & 36 \end{bmatrix}$$

$$bC = \begin{bmatrix} 0 & 14 & -21 \\ -7 & -49 & -28 \\ -21 & -35 & 63 \end{bmatrix}$$

$$aC + bC = \begin{bmatrix} 0 & 6 & -9 \\ -3 & -21 & -12 \\ -9 & -15 & -27 \end{bmatrix}$$

d. $a(B - C) = aB - aC$

$$B - C = \begin{bmatrix} 8 & -1 & -8 \\ -1 & -6 & -2 \\ 1 & -12 & -3 \end{bmatrix}$$

$$4(B - C) = \begin{bmatrix} 32 & -4 & -32 \\ -4 & -24 & -8 \\ 4 & -48 & -12 \end{bmatrix}$$

$$aB = \begin{bmatrix} 32 & -12 & -20 \\ 0 & 4 & 8 \\ 16 & -28 & 24 \end{bmatrix}$$

$$aC = \begin{bmatrix} 0 & -8 & 12 \\ 4 & 28 & 16 \\ 12 & 20 & 36 \end{bmatrix}$$

$$aB - aC =$$

$$\begin{bmatrix} 32 & -4 & -32 \\ -4 & -24 & -8 \\ 4 & -48 & -12 \end{bmatrix}$$

3. a $(A^T)^T = A$

$$A^T = \begin{bmatrix} 2 & 0 & -2 \\ -1 & 4 & 1 \\ 3 & 5 & 4 \end{bmatrix} \quad (A^T)^T = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ -2 & 1 & 4 \end{bmatrix} = A$$

b. $(A+B)^T = A^T + B^T$

$$A+B = \begin{bmatrix} 10 & -4 & -2 \\ 0 & 5 & 7 \\ 9 & -6 & 10 \end{bmatrix} \quad (A+B)^T = \begin{bmatrix} 10 & 0 & 2 \\ -4 & 5 & -6 \\ -9 & 7 & 10 \end{bmatrix} \quad B^T = \begin{bmatrix} 8 & 0 & 4 \\ -3 & 1 & -7 \\ -5 & 2 & 6 \end{bmatrix} \quad A^T + B^T = \begin{bmatrix} 10 & 0 & 2 \\ -4 & 5 & -6 \\ -9 & 7 & 10 \end{bmatrix}$$

c. $(aC)^T = aC^T$

$$\begin{bmatrix} 0 & -8 & 12 \\ 4 & 28 & 16 \\ 12 & 20 & 36 \end{bmatrix} \quad (aC)^T = \begin{bmatrix} 0 & 4 & 12 \\ -8 & 28 & 20 \\ 12 & 16 & 36 \end{bmatrix} \quad C^T = \begin{bmatrix} 0 & 1 & 3 \\ -2 & 7 & 5 \\ 3 & 4 & 9 \end{bmatrix} \quad 4C^T = \begin{bmatrix} 0 & 4 & 12 \\ -8 & 28 & 20 \\ 12 & 16 & 36 \end{bmatrix}$$

d. $(AB)^T = B^T A^T$

$$AB = \begin{bmatrix} 28 & -28 & 6 \\ 20 & -31 & 38 \\ 0 & -21 & 36 \end{bmatrix} \quad AB^T = \begin{bmatrix} 28 & 20 & 0 \\ -28 & -31 & -21 \\ 6 & 38 & 36 \end{bmatrix} \quad B^T A^T = \begin{bmatrix} 8 & 0 & 4 \\ -3 & 1 & -7 \\ -5 & 2 & 6 \end{bmatrix} \begin{bmatrix} 2 & 0 & -2 \\ -1 & 4 & 1 \\ 3 & 5 & 4 \end{bmatrix} = \begin{bmatrix} 28 & 20 & 0 \\ -28 & -31 & -21 \\ 6 & 38 & 36 \end{bmatrix}$$

5. $B = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$

$$B^{-1} = \frac{1}{20} \begin{bmatrix} 4 & 3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{20} & \frac{3}{20} \\ -\frac{4}{20} & \frac{2}{20} \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & \frac{3}{20} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix}$$

$$(2, 4) - (-3, 4)$$

$$\det B = 20$$

7. $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

$$D^{-1} = \frac{1}{6} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{6} & 0 \\ 0 & \frac{2}{6} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$$

$$\det D = 6$$

$$9. \begin{bmatrix} \frac{1}{2}(e^x + e^{-x}) & \frac{1}{2}(e^x - e^{-x}) \\ \frac{1}{2}(e^x - e^{-x}) & \frac{1}{2}(e^x + e^{-x}) \end{bmatrix}$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} \frac{1}{2}(e^x + e^{-x}) & -\frac{1}{2}(e^x - e^{-x}) \\ -\frac{1}{2}(e^x - e^{-x}) & \frac{1}{2}(e^x + e^{-x}) \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2}(e^x + e^{-x}) & -\frac{1}{2}(e^x - e^{-x}) \\ -\frac{1}{2}(e^x - e^{-x}) & \frac{1}{2}(e^x + e^{-x}) \end{bmatrix}$$

$$\det = \left(\frac{1}{2}(e^x + e^{-x}) \right) \left(\frac{1}{2}(e^x + e^{-x}) \right) - \left(\frac{1}{2}(e^x - e^{-x}) \right) \left(\frac{1}{2}(e^x - e^{-x}) \right)$$

$$= \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4}$$

$$= \frac{(e^x)^2 + 2e^x e^{-x} + (e^{-x})^2}{4} - \frac{(e^x)^2 - 2e^x e^{-x} + (e^{-x})^2}{4}$$

$$= \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4} = \frac{4}{4} = 1$$

$$11. B = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 2 & 4 \\ -3 & 4 \end{bmatrix}$$

$$(B^T)^{-1} = \frac{1}{20} \begin{bmatrix} 4 & -4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & -\frac{1}{5} \\ \frac{3}{20} & \frac{1}{10} \end{bmatrix}$$

$$(B^{-1})^T = \begin{bmatrix} \frac{1}{5} & -\frac{1}{5} \\ \frac{3}{20} & \frac{1}{10} \end{bmatrix}$$

$$\det B = (2 \cdot 4) - (-3 \cdot 4) = 20$$

$$\det B^T = (2 \cdot 4) - (-3 \cdot 4) = 20$$

$$B^{-1} = \frac{1}{20} \begin{bmatrix} 4 & 3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & \frac{3}{20} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix}$$

Exercício 1.5 - 10a20.

$$10. \begin{bmatrix} -3 & 6 \\ 4 & 5 \end{bmatrix} = \left[\begin{array}{cc|cc} -3 & 6 & 1 & 0 \\ 4 & 5 & 0 & 1 \end{array} \right] \xrightarrow{-\frac{1}{3}L_1} \left[\begin{array}{cc|cc} 1 & -2 & -\frac{1}{3} & 0 \\ 4 & 5 & 0 & 1 \end{array} \right] \xrightarrow{L_2 - 4L_1} \left[\begin{array}{cc|cc} 1 & -2 & -\frac{1}{3} & 0 \\ 0 & 13 & \frac{4}{3} & 1 \end{array} \right] \xrightarrow{\frac{1}{13}L_2}$$

$$\left[\begin{array}{cc|cc} 1 & -2 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{4}{39} & \frac{1}{13} \end{array} \right] \xrightarrow{L_1 + 2L_2} \left[\begin{array}{cc|cc} 1 & 0 & -\frac{5}{39} & \frac{2}{13} \\ 0 & 1 & \frac{4}{39} & \frac{1}{13} \end{array} \right] \quad A^{-1} = \begin{bmatrix} -\frac{5}{39} & \frac{2}{13} \\ \frac{4}{39} & \frac{1}{13} \end{bmatrix}$$

$$11. \begin{bmatrix} -1 & 3 \\ 3 & -2 \end{bmatrix} = \left[\begin{array}{cc|cc} -1 & 3 & 1 & 0 \\ 3 & -2 & 0 & 1 \end{array} \right] \xrightarrow{-L_1} \left[\begin{array}{cc|cc} 1 & -3 & -1 & 0 \\ 3 & -2 & 0 & 1 \end{array} \right] \xrightarrow{L_2 - 3L_1} \left[\begin{array}{cc|cc} 1 & -3 & -1 & 0 \\ 0 & 7 & 3 & 1 \end{array} \right] \xrightarrow{\frac{1}{7}L_2}$$

$$\left[\begin{array}{cc|cc} 1 & -3 & -1 & 0 \\ 0 & 1 & \frac{3}{7} & \frac{1}{7} \end{array} \right] \xrightarrow{L_1 + 3L_2} \left[\begin{array}{cc|cc} 1 & 0 & \frac{2}{7} & \frac{3}{7} \\ 0 & 1 & \frac{3}{7} & \frac{1}{7} \end{array} \right] \quad A^{-1} = \begin{bmatrix} \frac{2}{7} & \frac{3}{7} \\ \frac{3}{7} & \frac{1}{7} \end{bmatrix} \text{ ou } \frac{1}{7} \begin{bmatrix} 2 & 3 \\ 3 & 1 \end{bmatrix}$$

$$12. \begin{bmatrix} 6 & -4 \\ -3 & 2 \end{bmatrix} = \left[\begin{array}{cc|cc} 6 & -4 & 1 & 0 \\ -3 & 2 & 0 & 1 \end{array} \right] \xrightarrow{L_1 + 2L_2} \left[\begin{array}{cc|cc} 0 & 0 & 1 & 2 \\ -3 & 2 & 0 & 1 \end{array} \right] \quad \text{Não é inversível.}$$

$$13. \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix} = \left[\begin{array}{ccc|ccc} 3 & 4 & -1 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 2 & 5 & -4 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{L_2 \leftrightarrow L_1 \\ L_3 - L_1}} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 3 & 4 & -1 & 1 & 0 & 0 \\ 2 & 5 & -4 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{L_2 - 3L_1 \\ L_3 - 2L_1}} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 4 & -10 & 1 & -3 & 0 \\ 0 & 5 & -10 & 0 & -2 & 1 \end{array} \right]$$

$$\xrightarrow{L_3 - L_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 4 & -10 & 1 & -3 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \end{array} \right] \xrightarrow{\substack{L_2 \leftrightarrow L_3 \\ L_2 \cdot \frac{1}{4}}} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 4 & -10 & 1 & -3 & 0 \end{array} \right] \xrightarrow{L_3 - 4L_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & -10 & 5 & -7 & -4 \end{array} \right] \xrightarrow{\frac{1}{10}L_3}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{7}{10} & \frac{2}{5} \end{array} \right] \xrightarrow{L_1 - 3L_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{3}{2} & \frac{11}{10} & \frac{6}{5} \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{7}{10} & \frac{2}{5} \end{array} \right] \quad A^{-1} = \begin{bmatrix} -\frac{3}{2} & \frac{11}{10} & \frac{6}{5} \\ -1 & 1 & 1 \\ -\frac{1}{2} & \frac{7}{10} & \frac{2}{5} \end{bmatrix}$$

$$14. \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & | & 1 & 0 & 0 \\ 2 & 1 & 2 & | & 0 & 1 & 0 \\ 0 & 2 & 1 & | & 0 & 0 & 1 \end{bmatrix} \xrightarrow{L_2 - 2L_1} \begin{bmatrix} 1 & 2 & 0 & | & 1 & 0 & 0 \\ 0 & -3 & 2 & | & -2 & 1 & 0 \\ 0 & 2 & 1 & | & 0 & 0 & 1 \end{bmatrix} \xrightarrow{L_2 + L_3} \begin{bmatrix} 1 & 2 & 0 & | & 1 & 0 & 0 \\ 0 & -1 & 3 & | & -2 & 1 & 1 \\ 0 & 2 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{-1L_2} \begin{bmatrix} 1 & 2 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & -3 & | & 2 & -1 & -1 \\ 0 & 2 & 1 & | & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{L_1 - 2L_2 \\ L_3 - 2L_2}} \begin{bmatrix} 1 & 0 & 6 & | & -3 & 2 & 2 \\ 0 & 1 & -3 & | & 2 & -1 & -1 \\ 0 & 0 & 7 & | & -4 & 2 & 3 \end{bmatrix} \xrightarrow{\frac{1}{7}L_3} \begin{bmatrix} 1 & 0 & 6 & | & -3 & 2 & 2 \\ 0 & 1 & -3 & | & 2 & -1 & -1 \\ 0 & 0 & 1 & | & -4/7 & 2/7 & 3/7 \end{bmatrix} \xrightarrow{\substack{L_1 - 6L_3 \\ L_2 + 3L_3}}$$

$$\begin{bmatrix} 1 & 0 & 0 & | & 3/7 & 2/7 & -4/7 \\ 0 & 1 & 0 & | & 2/7 & -1/7 & 2/7 \\ 0 & 0 & 1 & | & -4/7 & 2/7 & 3/7 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3/7 & 2/7 & -4/7 \\ 2/7 & -1/7 & 2/7 \\ -4/7 & 2/7 & 3/7 \end{bmatrix} \text{ ou } \frac{1}{7} \begin{bmatrix} 3 & 2 & -4 \\ 2 & -1 & 2 \\ -4 & 2 & 3 \end{bmatrix}$$

$$15. \begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 2 & 4 & 1 & | & 0 & 1 & 0 \\ -4 & 2 & -9 & | & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-L_1} \begin{bmatrix} 1 & -3 & 4 & | & -1 & 0 & 0 \\ 2 & 4 & 1 & | & 0 & 1 & 0 \\ -4 & 2 & -9 & | & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{L_2 - 2L_1 \\ L_3 + 4L_1}} \begin{bmatrix} 1 & -3 & 4 & | & -1 & 0 & 0 \\ 0 & 10 & -7 & | & 2 & 1 & 0 \\ 0 & -10 & 7 & | & -4 & 0 & 1 \end{bmatrix} \xrightarrow{L_3 + L_2}$$

$$\begin{bmatrix} 1 & -3 & 4 & | & -1 & 0 & 0 \\ 0 & 10 & -7 & | & 2 & 1 & 0 \\ 0 & 0 & 0 & | & -2 & 1 & 1 \end{bmatrix}$$

Não possui inversa

$$16. A \cdot 10 = \begin{bmatrix} 2 & 2 & -4 & | & 10 & 0 & 0 \\ 2 & 2 & 1 & | & 0 & 10 & 0 \\ 2 & -8 & 1 & | & 0 & 0 & 10 \end{bmatrix} \xrightarrow{\substack{L_2 - L_1 \\ L_3 - L_1}} \begin{bmatrix} 2 & 2 & -4 & | & 10 & 0 & 0 \\ 0 & 0 & 5 & | & -10 & 10 & 0 \\ 0 & -10 & 5 & | & -10 & 0 & 10 \end{bmatrix} \xrightarrow{\substack{L_2 \leftrightarrow L_3}} \begin{bmatrix} 2 & 2 & -4 & | & 10 & 0 & 0 \\ 0 & -10 & 5 & | & -10 & 0 & 10 \\ 0 & 0 & 5 & | & -10 & 10 & 0 \end{bmatrix}$$

$$\xrightarrow{\substack{\frac{1}{2}L_1 \\ -\frac{1}{10}L_2}} \begin{bmatrix} 1 & 1 & -2 & | & 5 & 0 & 0 \\ 0 & 1 & -1/2 & | & 1 & 0 & -1 \\ 0 & 0 & 1 & | & -2 & 2 & 0 \end{bmatrix} \xrightarrow{\substack{L_2 + \frac{1}{2}L_3 \\ L_1 + 2L_3}} \begin{bmatrix} 1 & 1 & 0 & | & 1 & 4 & 0 \\ 0 & 1 & 0 & | & 0 & 1 & -1 \\ 0 & 0 & 1 & | & -2 & 2 & 0 \end{bmatrix} \xrightarrow{L_1 - L_2} \begin{bmatrix} 1 & 0 & 0 & | & 1 & 3 & 1 \\ 0 & 1 & 0 & | & 0 & 1 & -1 \\ 0 & 0 & 1 & | & -2 & 2 & 0 \end{bmatrix}$$

$$\xrightarrow{\frac{1}{5}L_3}$$

$$A^{-1} = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & -1 \\ -2 & 2 & 0 \end{bmatrix}$$

$$17. \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{L_3 - L_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -1 & 0 & 1 \end{array} \right] \xrightarrow{L_3 - L_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array} \right] \xrightarrow{-\frac{1}{2}L_3}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right] \xrightarrow{\substack{L_1 - L_3 \\ L_2 - L_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \text{ or } \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

$$18. \left[\begin{array}{ccc|ccc} \sqrt{2} & 3\sqrt{2} & 0 & 1 & 0 & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{L_1 \cdot \frac{1}{\sqrt{2}}} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 \\ -4 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{L_2 + (4\sqrt{2}L_1)} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 13\sqrt{2} & 0 & 4 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{L_2 \\ \frac{1}{13\sqrt{2}}}}$$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 1 & 0 & \frac{4\sqrt{2}}{13} & \frac{1}{13\sqrt{2}} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{L_1 - 3L_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{\sqrt{2}}{26} & -\frac{3\sqrt{2}}{26} & 0 \\ 0 & 1 & 0 & \frac{4\sqrt{2}}{13} & \frac{1}{13\sqrt{2}} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} \frac{\sqrt{2}}{26} & -\frac{3\sqrt{2}}{26} & 0 \\ \frac{4\sqrt{2}}{13} & \frac{1}{13\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$19. \left[\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 2 & 7 & 6 & 0 & 1 & 0 \\ 2 & 7 & 7 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{L_2 - L_1 \\ L_3 - L_1}} \left[\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array} \right] \xrightarrow{L_3 - L_2} \left[\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right] \xrightarrow{\substack{L_1 - 6L_2 \\ L_1 - 6L_3}}$$

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 7 & 0 & -6 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right] \xrightarrow{\frac{1}{2}L_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{7}{2} & 0 & -3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} \frac{7}{2} & 0 & -3 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

20.
$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 5 & 5 & 0 & 0 & 0 & 1 & 0 \\ 1 & 3 & 5 & 7 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{L_2-L_1 \\ L_3-L_1 \\ L_4-L_1}} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 5 & 5 & 0 & -1 & 0 & 1 & 0 \\ 0 & 3 & 5 & 7 & -1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{L_3-L_2 \\ L_4-L_2}} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 5 & 7 & 0 & 0 & -1 & 1 \end{array} \right]$$

$$\xrightarrow{L_4-L_3} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 7 & 0 & 0 & -1 & 1 \end{array} \right] \xrightarrow{\substack{\frac{1}{3}L_2 \\ \frac{1}{5}L_3 \\ \frac{1}{7}L_4}} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -\frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{7} & \frac{1}{7} \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & -\frac{1}{7} & \frac{1}{7} \end{bmatrix}$$

Exercício 2.1 - (1, 3, 5, 8, 19, 21, 25).

1- $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 7 & -1 \\ -3 & 1 & 4 \end{bmatrix}$

$1^a L = M_{11} = \begin{vmatrix} 7 & -1 \\ 1 & 4 \end{vmatrix} = 7 \cdot 4 - (-1 \cdot 1) = 29 \quad C_{11} = (-1)^{1+1} \cdot 29 = 29$

$M_{12} = \begin{vmatrix} 6 & -1 \\ -3 & 4 \end{vmatrix} = 21 \quad C_{12} = (-1)^{1+2} \cdot 21 = -21$

$M_{13} = \begin{vmatrix} 6 & 7 \\ -3 & 1 \end{vmatrix} = 27 \quad C_{13} = (-1)^{1+3} \cdot 27 = 27$

$2^a L = M_{21} = \begin{vmatrix} -2 & 3 \\ 1 & 4 \end{vmatrix} = -11 \quad C_{21} = (-1)^{2+1} \cdot -11 = 11$

$M_{22} = \begin{vmatrix} 1 & 3 \\ -3 & 4 \end{vmatrix} = 13 \quad C_{22} = (-1)^{2+2} \cdot 13 = 13$

$M_{23} = \begin{vmatrix} 1 & -2 \\ -3 & 1 \end{vmatrix} = -5 \quad C_{23} = (-1)^{2+3} \cdot -5 = 5$

$3^a L = M_{31} = \begin{vmatrix} -2 & 3 \\ 7 & -1 \end{vmatrix} = -19 \quad C_{31} = (-1)^{3+1} \cdot -19 = -19$

$M_{32} = \begin{vmatrix} 1 & 3 \\ 6 & -1 \end{vmatrix} = -19 \quad C_{32} = (-1)^{3+2} \cdot -19 = 19$

$M_{33} = \begin{vmatrix} 1 & -2 \\ 6 & 7 \end{vmatrix} = 19 \quad C_{33} = (-1)^{3+3} \cdot 19 = 19$

3- $A = \begin{bmatrix} 4 & -1 & 16 \\ 0 & 0 & -33 \\ 4 & 1 & 014 \\ 4 & 1 & 32 \end{bmatrix}$ a. $M_{13} = \begin{bmatrix} 0 & 0 & 3 \\ 4 & 1 & 14 \\ 4 & 1 & 2 \end{bmatrix} = 3(4 \cdot 1 - 4 \cdot 1) = 0 \quad C_{13} = (-1)^{1+3} \cdot 0 = 0$

b. $M_{23} = \begin{bmatrix} 4 & -1 & 6 \\ 4 & 1 & 14 \\ 4 & 1 & 2 \end{bmatrix} \begin{matrix} 4 & -1 \\ 4 & 1 \\ 4 & 1 \end{matrix} = \begin{matrix} 4 \cdot 1 \cdot 2 + (-1) \cdot 14 \cdot 4 + 6 \cdot 4 \cdot 1 - (-1) \cdot 4 \cdot 2 - 4 \cdot 4 \cdot 1 - 6 \cdot 1 \cdot 4 \\ 8 - 56 + 24 + 8 - 56 - 24 \\ -96 \end{matrix}$

$C_{23} = (-1)^{2+3} \cdot -96 = 96$

c. $M_{22} = \begin{bmatrix} 4 & 1 & 6 \\ 4 & 0 & 14 \\ 4 & 3 & 2 \end{bmatrix} \begin{matrix} 4 & 1 \\ 4 & 0 \\ 4 & 3 \end{matrix} = \begin{matrix} 0 + 56 + 72 - 8 - 168 + 0 \\ -48 \end{matrix}$

$C_{22} = (-1)^{2+2} \cdot -48 = -48$

d. $M_{31} = \begin{bmatrix} -1 & 1 & 6 \\ 1 & 0 & 14 \\ 1 & 3 & 2 \end{bmatrix} \begin{matrix} -1 & 1 \\ 1 & 0 \\ 1 & 3 \end{matrix} = \begin{matrix} 0 + 14 + 18 - 2 + 42 + 0 \\ 72 \end{matrix}$

$C_{31} = (-1)^{3+1} \cdot 72 = -72$

$$5. \begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} = (3 \cdot 4) - (5 \cdot -2) = 12 - (-10) = 22$$

$$A^{-1} = \frac{1}{22} \begin{bmatrix} 4 & -5 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} \frac{4}{22} & -\frac{5}{22} \\ \frac{2}{22} & \frac{3}{22} \end{bmatrix}$$

$$8. \begin{bmatrix} \sqrt{2} & \sqrt{6} \\ 4 & \sqrt{3} \end{bmatrix} = \sqrt{2} \cdot \sqrt{3} - (\sqrt{6} \cdot 4) = \sqrt{6} - 4\sqrt{6} = -3\sqrt{6}$$

$$A^{-1} = \frac{1}{-3\sqrt{6}} \begin{bmatrix} \sqrt{3} & -\sqrt{6} \\ -4 & \sqrt{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{-3\sqrt{6}} & \frac{-\sqrt{6}}{-3\sqrt{6}} \\ \frac{-4}{-3\sqrt{6}} & \frac{\sqrt{2}}{-3\sqrt{6}} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{3}}{3} & \frac{1}{3} \\ \frac{2\sqrt{6}}{9} & -\frac{\sqrt{2}}{9} \end{bmatrix}$$

$$19. a. \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 5 \\ 1 & 9 & -4 \end{bmatrix} = 3 \cdot \begin{vmatrix} -1 & 5 \\ 9 & -4 \end{vmatrix} - 0 + 0 = 3[-1 \cdot -4 - 5 \cdot 9] = -123$$

$$b. 3 \begin{vmatrix} -1 & 5 \\ 9 & -4 \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 \\ 9 & -4 \end{vmatrix} + 1 \begin{vmatrix} 0 & 0 \\ -1 & 5 \end{vmatrix} = 3 \cdot -41 - 0 + 1 \cdot 0 = -123$$

$$c. -2 \begin{vmatrix} 0 & 0 \\ 9 & -4 \end{vmatrix} + (-1) \begin{vmatrix} 3 & 0 \\ 1 & -4 \end{vmatrix} - 5 \begin{vmatrix} 3 & 0 \\ 1 & 9 \end{vmatrix} = +12 - 135 = -123$$

$$d. 0 + (-1) \begin{vmatrix} 3 & 0 \\ 1 & -4 \end{vmatrix} - 9 \begin{vmatrix} 3 & 0 \\ 2 & 5 \end{vmatrix} = 12 - 135 = -123$$

$$e. -1 \begin{vmatrix} 0 & 0 \\ -1 & 5 \end{vmatrix} - 9 \begin{vmatrix} 3 & 0 \\ 2 & 5 \end{vmatrix} + (-4) \begin{vmatrix} 3 & 0 \\ 9 & -1 \end{vmatrix} = -135 + 12 = -123$$

$$f. 0 - 5 \begin{vmatrix} 3 & 0 \\ 1 & 9 \end{vmatrix} + (-4) \begin{vmatrix} 3 & 0 \\ 2 & -1 \end{vmatrix} = -135 + 12 = -123$$

$$21. \begin{bmatrix} -3 & 0 & 7 \\ 2 & 5 & 1 \\ -1 & 0 & 5 \end{bmatrix} \quad \det A = a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32}$$

$$\det A = 0 + 5 \cdot (-1)^{2+2} \cdot 8 + 0$$

$$\det A = -40$$

$$a_{12} = 0$$

$$a_{22} = \begin{bmatrix} -3 & 7 \\ -1 & 5 \end{bmatrix} = (-3 \cdot 5) - (7 \cdot -1) = -8$$

$$a_{32} = 0$$

$$25. \begin{bmatrix} 3 & 3 & 0 & 5 \\ 2 & 2 & 0 & -2 \\ 4 & 1 & -3 & 0 \\ 8 & 10 & 3 & 0 \end{bmatrix} \quad \det A = 0 \cdot C_{13} + 0 \cdot C_{23} + (-3) \cdot C_{33} + 3 \cdot C_{43}$$

$$\det A = (-3 \cdot 1 \cdot 32) + (3 \cdot -1 \cdot -60)$$

$$\det A = -96 + 180$$

$$\det A = 84$$

$$M_{33} = \begin{bmatrix} 3 & 3 & 5 \\ 2 & 2 & -2 \\ 2 & 10 & 3 \end{bmatrix} = [3 \cdot 2 \cdot 3 + 3 \cdot -2 \cdot 2 + 5 \cdot 2 \cdot 10] - [5 \cdot 2 \cdot 2 + 3 \cdot 2 \cdot 10 + 3 \cdot -2 \cdot 2] = 32$$

$$M_{43} = \begin{bmatrix} 3 & 3 & 5 \\ 2 & 2 & -2 \\ 4 & 1 & 0 \end{bmatrix} = [0 \cdot -24 + 10] - [40 + 6 \cdot 0] = -60$$

Ejercicio 2.3 - (1, 3, 5, 7, 11, 21, 26, 30).

1. $A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}; K=2$

$$K^m \det A = K^2 \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix} = 2^2 \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$KA = \begin{bmatrix} -2 & 4 \\ 6 & 8 \end{bmatrix} = (8-24) = -16-24 = -40$$

$$4 \cdot ((4 \cdot -1) - (2 \cdot 3))$$

$$4 \cdot -10$$

$$K^m \det A = -40$$

3. $A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix}, K=-2$

$$K^m \det A = (-2)^3 \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix} = -8 \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix} =$$

$$KA = \begin{bmatrix} -4 & 2 & -6 \\ -6 & -4 & -2 \\ -2 & -8 & -10 \end{bmatrix} = \begin{bmatrix} -4 & 2 \\ -6 & -4 \\ -2 & -8 \end{bmatrix} = (-160 + 8 - 288) - (120 - 64 - 48)$$

$$\det(KA) = -448$$

$$\begin{bmatrix} 2 & -1 & 3 & 2 & -1 \\ 3 & 2 & 1 & 3 & 2 \\ 1 & 4 & 5 & 1 & 4 \end{bmatrix} = \begin{matrix} 20-1+36-(-15+8+6) \\ 55+1 \\ 56 \end{matrix}$$

$$K^m \det A = -8 \cdot 56 = -448$$

5. $A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}_{3 \times 3}$

$$\det AB = \begin{bmatrix} 2 \cdot 1 + 1 \cdot 7 + 0 \cdot 5 & 2 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 2 \cdot 2 + 1 \cdot 2 + 0 \cdot 1 \\ 3 \cdot 1 + 4 \cdot 7 + 0 \cdot 5 & 3 \cdot 1 + 4 \cdot 1 + 0 \cdot 0 & 3 \cdot 2 + 4 \cdot 2 + 0 \cdot 1 \\ 0 \cdot 1 + 0 \cdot 7 + 2 \cdot 5 & 0 \cdot 1 + 0 \cdot 1 + 2 \cdot 0 & 0 \cdot 2 + 0 \cdot 2 + 2 \cdot 1 \end{bmatrix}$$

$B = \begin{bmatrix} 1 & -1 & 3 \\ 7 & 1 & 2 \\ 5 & 0 & 1 \end{bmatrix}_{3 \times 3}$

$$\begin{bmatrix} 9 & -1 & 4 \\ 31 & +1 & 12 \\ 10 & 2 & 2 \end{bmatrix} = (10 \cdot 17) = -170$$

$$\det BA = -17 \cdot 10 = -170$$

$$A+B = \begin{bmatrix} 3 & 0 & 3 \\ 10 & 5 & 2 \\ 5 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 10 & 5 \\ 5 & 0 \end{bmatrix} = (45+0+0) - (0+0+75) = -30$$

$$A = \begin{bmatrix} 2 & 1 & 0 & 2 & 1 \\ 3 & 4 & 0 & 3 & 4 \\ 0 & 0 & 2 & 0 & 0 \end{bmatrix} = (16+0+0) - (6+0+0) = 10$$

$$B = \begin{bmatrix} 1 & -1 & 3 & 1 & -1 \\ 7 & 1 & 2 & 7 & 1 \\ 5 & 0 & 1 & 5 & 0 \end{bmatrix} = (1-10+0) - (-7+0+15) = -17$$

$$\det A + \det B = 10 - 17 = -7$$

$$7. A = \begin{bmatrix} 2 & 5 & 5 \\ -1 & 4 & 0 \\ 0 & 4 & 3 \end{bmatrix} \begin{array}{l} 2 \quad 5 \\ -1 \quad -1 \\ 2 \quad 4 \end{array} = (-6 + 0 - 20) - (-10 + 0 - 10) \\ = -26 + 20 \\ \det A = -6$$

É invertível

$$11. A = \begin{bmatrix} 4 & 2 & 9 \\ -2 & 1 & -4 \\ 3 & 1 & 6 \end{bmatrix} \begin{array}{l} 4 \quad 2 \\ -2 \quad 1 \\ 3 \quad 1 \end{array} = 24 - 24 - 16 - (-24 - 16 + 24) \\ = -16 + 16 \\ \det A = 0$$

Não é invertível

$$21. A = \begin{bmatrix} 2 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{bmatrix} \det A = 4$$

$$\begin{array}{lll} C_{11} = 12 - (-20) = 32 & C_{12} = -6 & C_{13} = 4 \\ C_{21} = 0 & C_{22} = 4 & C_{23} = -6 \\ C_{31} = 0 & C_{32} = 0 & C_{33} = 2 \end{array}$$

$$C = \begin{bmatrix} 32 & 0 & 0 \\ -6 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad \text{adj} = \begin{bmatrix} 32 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad A^{-1} = \frac{1}{4} \cdot \begin{bmatrix} 32 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/2 \end{bmatrix}$$

$$26. \begin{cases} x - 4y + z = 6 \\ 4x - y + 2z = -1 \\ 2x + 2y - 3z = -20 \end{cases}$$

$$A = \begin{bmatrix} 1 & -4 & 1 \\ 4 & -1 & 2 \\ 2 & 2 & -3 \end{bmatrix} \quad A_1 = \begin{bmatrix} 6 & -4 & 1 \\ -1 & -1 & 2 \\ -20 & 2 & -3 \end{bmatrix} \quad A_2 = \begin{bmatrix} 1 & 6 & 1 \\ 4 & -1 & 2 \\ 2 & -20 & -3 \end{bmatrix} \quad A_3 = \begin{bmatrix} 1 & -4 & 6 \\ 4 & -1 & -1 \\ 2 & 2 & -20 \end{bmatrix}$$

$$\det A = 3 - 16 + 8 - (48 + 4 - 2) = -55 \quad \det A_1 = 18 + 160 - 20 - 24 + 12 = 144 \quad \det A_2 = 30 - 90 + 40 + 20 = -20 \quad \det A_3 = 20 + 20 + 40 + 20 = 100$$

$$x = \frac{144}{-55} \quad y = \frac{61}{-55} \quad z = \frac{230}{-55}$$

$$30. A = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{l} C_{11} = \cos \theta \cdot 1 \\ C_{12} = \sin \theta \cdot (-1) \\ C_{13} = 0 \end{array}$$

$$C = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Ad} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det A = 1 \cdot \begin{vmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{vmatrix}$$

$$\det A = \cos^2 \theta + \sin^2 \theta$$

$$\det A = 1$$

$$\begin{array}{l} C_{21} = \sin \theta \cdot (-1) \\ C_{22} = \cos \theta \cdot 1 \\ C_{23} = 0 \\ C_{31} = 0 \\ C_{32} = 0 \\ C_{33} = 1 \end{array}$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$